

1. With a relatively small set of hyperparameters, it is OK to use a grid search. True/False? 1 point
- ☒ True
- ☐ False
2. If it is only possible to tune two parameters from the following due to limited computational resources. Which two would you choose? 1 point
- ☒ The β parameter of the momentum in gradient descent.
- ☐ β_1, β_2 in Adam.
- ☐ ϵ in Adam.
- ☒ α
3. Using the "Panda" strategy, it is possible to create several models. True/False? 1 point
- ☐ False
- ☒ True
4. Knowing that the hyperparameter α should be in the range of 0.001 and 1.0. Which of the following is the recommended way to sample a value for α ? 1 point
- ☐
- `r = -5*np.random.rand()`
`alpha = 10**r`
- ☐
- `r = np.random.rand()`
`alpha = 0.001 + r*0.999`
- ☐
- `r = 4*np.random.rand()`
`alpha = 10**r`
- ☒
- `r = -3*np.random.rand()`
`alpha = 10**r`
5. Finding good hyperparameter values is very time-consuming. So typically you should do it once at the start of the project, and try to find very good hyperparameters so that you don't ever have to tune them again. True or false? 1 point
- ☐ True
- ☒ False
6. When using batch normalization it is OK to drop the parameter $W^{[l]}$ from the forward propagation since it will be subtracted out when we compute $\hat{x}^{[l]} = \gamma z_{\text{normalise}}^{[l]} + \beta^{[l]}$. True/False? 1 point
- ☐ True
- ☒ False
7. When using normalization:
- $$z_{\text{norm}}^{(i)} = \frac{z^{(i)} - \mu}{\sqrt{\sigma^2 + \epsilon}}$$
- In case σ is too small, the normalization of $z^{(i)}$ may fail since division by 0 may be produced due to rounding errors. True/False? 1 point
- ☒ False
- ☐ True
8. Which of the following are true about batch normalization? 1 point
- ☒ When using batch normalization we introduce two new parameters $\gamma^{[l]}, \beta^{[l]}$ that must be "learned" or trained.
- ☒ The parameters $\gamma^{[l]}$ and $\beta^{[l]}$ set the variance and mean of $\hat{x}^{[l]}$.
- ☒ $z_{\text{norm}}^{(i)} = \frac{z^{(i)} - \mu}{\sqrt{\sigma^2}}$.
- ☐ $\beta^{[l]}$ and $\gamma^{[l]}$ are hyperparameters that must be tuned by random sampling in a logarithmic scale.
9. A neural network is trained with Batch Norm. At test time, to evaluate the neural network on a new example you should perform the normalization using μ and σ^2 estimated using an exponentially weighted average across mini-batches seen during training. True/false? 1 point
- ☐ False
- ☒ True
10. Which of the following are some recommended criteria to choose a deep learning framework? 1 point
- ☐ It must use Python as the primary language.
- ☒ Running speed.
- ☐ It must run exclusively on cloud services, to ensure its robustness.
- ☐ It must be implemented in C to be faster.